

Later improvement. The signed-target compression below remains an input to `factorial_order_eventual_conductor.tex`, which improves the eventual-size threshold to $\exp(O(n \log n))$ using a new prefix completion. The older conductor estimate in this note is retained as an independently proved bound.

An explicit threshold for eventual equal-sided fair sizes

Quantitative companion to the positive-completion theorem

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The qualitative positive-completion and Ore argument is due to the parallel research note in `../positive_saturation/`. This companion extracts bounds from that argument and from `signed_fair_signatures.tex`. An independent mathematical review of both the majorant and the base-digit compression is complete; this is not machine-checked formal verification.

Theorem 1. Put $P = \prod_{p \leq n} p$, $Q = (n!)^{n-1}$, and, for $n \geq 2$, define

$$L_* = 2^{2n+3} n^4 n! P, \quad q = \lceil \log_2 L_* \rceil, \quad B_n = Q 2^{nq}.$$

Every positive integer M divisible by P and satisfying $M \geq B_n^2/P$ is the common number of faces of an n -die permutation-fair collection. In particular there is a threshold

$$M_0(n) \leq \exp(O(n^2 \log n))$$

above which subset-factorial divisibility is the only obstruction to equal-sided sizes.

This is a bound on the threshold beyond which *every* admissible size occurs. It does not improve the smallest-size upper bound $2^{25}(n!)^2$.

Use the reduced squarefree associative algebra from the signed-signature note, and write $T_a = \exp(a \sum_i X_i)$. A positive word has ordinary length equal to its total number of letters.

Lemma 2 (Quantitative positive completion). *Every positive word u of length at most H , where $H \geq 1$, is a prefix of a positive word with signature T_A , for some integer $1 \leq A \leq QH$.*

Proof. First append letters until all counts are equal to $a = \max(1, \max_i c_i(u)) \leq H$. If the logarithm is $a \sum_i X_i + D_k + \text{higher degrees}$, concatenate all $n!$ relabelings, placing the identity relabeling first. The degree- k term is $\sum_{\sigma \in S_n} \sigma D_k$, since the common degree-one parts commute and their brackets with D_k have degree at least $k + 1$. This sum vanishes: an invariant multilinear Lie polynomial of degree $k > 1$ is zero. Indeed, on each k -letter support invariance forces all associative-word coefficients to be equal, whereas abelianization of a Lie polynomial of degree greater than one vanishes. Concatenation thus kills the degree- k error and multiplies all counts by $n!$. Apply this for $k = 2, \dots, n$, including a harmless averaging step when the current error vanishes. The original word remains a prefix, and the final common count is exactly $A = Qa$. $\square$

Lemma 3 (Short common right multiples: Mal'cev's identity). *Two positive words of length at most $H \geq 1$ have a common right multiple of length at most $2^n H$. Each of the two right complements can be chosen with length at most $(2^n - 1)H$.*

Proof. The group of reduced signatures is nilpotent of class at most n : if $\mathfrak{m}$ is the positive-degree ideal, then $[1 + \mathfrak{m}^a, 1 + \mathfrak{m}^b] \subseteq 1 + \mathfrak{m}^{a+b}$ and $\mathfrak{m}^{n+1} = 0$. Given words a, b , define positive words

$$A_0 = a, \quad B_0 = b, \quad A_{j+1} = A_j B_j, \quad B_{j+1} = B_j A_j.$$

Writing $g_j = A_j B_j^{-1}$ in the group, one has

$$g_{j+1} = [A_j, B_j] = [g_j, B_j].$$

Inductively $g_j \in 1 + \mathfrak{m}^{j+1}$, so $A_n = B_n$ as signatures. Both contain exactly 2^n original blocks; they begin with a and b , respectively. Removing those first blocks gives the stated positive complements.

This is the classical positive nilpotency identity of Mal'cev, recalled in [1, §5.1]; its use here requires no general semigroup theorem. $\square$

Lemma 4 (Short right fractions). *A signed word of length $L \geq 1$ can be written as uv^{-1} , with u, v positive words of length at most 2^{nq} , where*

$$q = \lceil \log_2 L \rceil.$$

Consequently its group element g satisfies $gT_A \in \mathcal{M}_+$ for some positive fair size $A \leq Q2^{nq}$, where $\mathcal{M}_+$ is the monoid of signatures of positive words.

Proof. Each signed generator has a right-fraction representation whose numerator and denominator have length at most one. Suppose two fractions $u_1v_1^{-1}$ and $u_2v_2^{-1}$ have all four lengths at most H . Lemma 3 gives $v_1a = u_2b$ with complements of length at most $(2^n - 1)H$. Their product is

$$(u_1a)(v_2b)^{-1}.$$

Both new lengths are at most $2^n H$. A balanced binary tree, padded by empty fractions if necessary, has at most q levels and proves the bound. Finally complete $vc = T_A$ by Lemma 2; then $gT_A = uc$ is positive and $A \leq Q2^{nq}$. $\square$

Lemma 5 (A short signed fair target). *The integral target T_P has a signed-word representative of length at most L_* .*

Proof. The signed-signature theorem applies because P satisfies every subset-factorial divisibility condition. We bound its constructive degree-correction proof. Start with $g_1 = \prod_i (1 + X_i)^P$, of length nP . Write $A = 2nP$. For the correction in degree k , let $b_{S,\pi}$ be the integer coefficients in the fixed-last-letter multilinear Lie basis, and put

$$c_k = 2^{k-1} \sum_{S,\pi} |b_{S,\pi}|.$$

Each basis bracket has associative coefficient ℓ^1 norm 2^{k-1} . Its coordinate is a coefficient of a distinct associative word, so the sum of absolute basis coordinates is no larger than the coefficient ℓ^1 norm of the degree- k error.

Products of correction factors and their inverses are coefficientwise majorized, in these norms, by $\exp(\sum c_j x^j)$. Both the initial inverse g_1^{-1} and the target T_P are majorized by $\exp(nPx)$. Therefore

$$c_k \leq 2^{k-1} [x^k] \exp \left(Ax + \sum_{2 \leq j < k} c_j x^j \right).$$

Inductively,

$$c_k \leq 2^{k^2} A^k. \quad (*)$$

To check this, normalize $d_j = c_j/A^j$, set $d_1 = 1$, and expand the exponential over partitions of k into parts less than k . There are at most 2^{k-1} partitions. In each one, $\sum_j j^2 m_j \leq (k-1)^2 + 1 = k^2 - 2k + 2$, since it has at least two positive parts. The factorial denominators $\prod m_j!$ only decrease the bound. Thus the recurrence is at most $2^{k-1} 2^{k-1} 2^{k^2-2k+2} = 2^{k^2}$, proving $(*)$.

It remains to represent a potentially large coefficient efficiently. For a coefficient $b = b_{S,\pi}$ choose the integer $R = 2^k A + 1$, so $|b| < R^k$ by (*). Write $|b| = \sum_{j=0}^{k-1} d_j R^j$ in base R . Each nonzero term $d_j R^j$ is a product of k positive integers at most R : use d_j , j copies of R , and $k - j - 1$ copies of 1. Put the sign of b into the first factor.

Replace the k generators in a nested group commutator by those signed integer powers. The result is exactly $1 + \text{sgn}(b) d_j R^j B_{S,\pi}$, since repeated-letter products vanish. Its signed-word length is at most $(3 \cdot 2^{k-1} - 2)R \leq k 2^k R$. Multiplying the at most k digit factors adds their coefficients, because $B_{S,\pi}^2 = 0$. Thus $1 + b B_{S,\pi}$ costs at most $k^2 2^k R \leq k^2 2^{2k+1} A$ letters.

There are $\binom{n}{k} (k-1)! \leq n!$ coordinates in degree k . The complete signed word therefore has length

$$L \leq nP + \sum_{k=2}^n n! k^2 2^{2k+1} A \leq 2^{2n+3} n^4 n! P = L_*.$$

□

Proof of Theorem 1. Apply Lemmas 4 and 5 to $g = T_P$. There is a positive fair word with common count $A \leq B_n$, and $T_P T_A = T_{A+P}$ also has a positive representative. Arithmetic necessity implies $P \mid A$. Set $a = A/P$. Positive concatenation therefore realizes every nonnegative integer combination of a and $a+1$, multiplied by P . Every integer $t \geq a(a-1)$ is such a combination: write $t = qa + r$, $0 \leq r < a$; then $q \geq a-1 \geq r$ and $t = (q-r)a + r(a+1)$. Consequently every admissible size at least $A(A-P)/P \leq B_n^2/P$ is realized.

Finally even the elementary bound $P \leq n!$ gives $\log L_* = O(n \log n)$, while $\log Q = O(n^2 \log n)$. The definition of B_n yields $\log B_n = O(n^2 \log n)$, as asserted. □

References

- [1] R. I. Grigorchuk, *On Growth in Group Theory*, Proceedings of the International Congress of Mathematicians, Kyoto 1990, Vol. I, Mathematical Society of Japan, 1991, pp. 325–338, especially §5.1, printed p. 331. Primary proceedings scan. The positive identity is attributed there to A. I. Mal'cev, *Nilpotent semigroups*, Uchen. Zap. Ivanovsk. Ped. In-ta 4 (1953), 107–111.